College of Information Technology and Computing
Department of Information Technology

SYLLABUS
Course Title: **IT ELECTIVE 3(Decision Support Systems Technologies)**
Course Code: IT414
Credits: 3 units (2 hours lecture, 3 hours laboratory)

USTP Vision

A nationally-recognized Science and Technology (S&T) university providing the vital link between education and the economy

USTP Mission

- Bring the world of work (industry) into the actual higher education and training of the students;
- Offer entrepreneurs of the opportunity to maximize their business potentials through a gamut of services from product conceptualization to commercialization;
- Contribute significantly to the national development goals of food security and

Semester/Year: 1st Semester, SY 2025-2026 Class Schedule: Bldg./Rm. No. IT414 – 4R8 – Lecture(T7:00-11:00AM), Laboratory(Th7:00-10:00AM rm09304) IT414 – 4R9 - Lecture(M1:00- 3:00PM), Laboratory(T7:00-10:00AM 09305) IT414 – 4R10 - Lecture(M4:00- 6:00PM), Laboratory(Th7:00-10:00AM 09302) IT414 – 4R11 - Lecture(T7:00-11:00 AM), Laboratory(Th10:00- 1:00 PM 09304) IT414 – 4R12 - Lecture(M 8:00-10:00 AM), Laboratory(T10:00- 1:00 PM 09305)	Prerequisite(s): **IT325 – IT Elective 2 (Data Mining)** Co-requisite(s): **none**
Instructor: Love Jhoye M. Raboy, PhD (IT414 – 4R8, IT414 – 4R11) Loredel B. Tamayo (IT414-4R10) Charlene B. Vallar, MIT (IT414-4R9, IT414-4R12) Email: lovejhoye.raboy@ustp.edu.ph loredel.tamayo@ustp.edu.ph charlane.vallar@ustp.edu.ph Mobile No.: +63.88.856.1738 Local 1153	Consultation Schedule: Love Jhoye Raboy (M1:00-3:00pm, T11:00-12:00nn. M 1:00-3:00PM) Loredel Tamayo (Th 10:00-12:00nn) Charlane Vallar ( ) Bldg.Rm. No.: IT Faculty Room, 4th Floor, ICT bldg., USTP Office Phone No./Local: +63.88.856.1738 Local 1153

I. Course Description: *[Why does this course exist? How does it fit in with the rest of the field/area's curriculum?]*
Decision Support Systems (DSS) are tools that decision makers use to gain a better understanding of their business. Modern-day DSS incorporate theories and technologies used for business analytics, business intelligence, data mining, and advanced techniques such as machine learning and predictive analytics. This course covers key topics in these areas, with a focus on applying machine learning and predictive analytics to support decision-making. Students will also explore text mining and deep learning that can enhance DSS capabilities, providing the necessary skills to support decision makers in organizations.

energy sufficiency through technology solutions.

Program Educational Objectives:

PEO1: Proficient in the IT field and able to engage constantly in technological and professional advancement by pursuing a higher academic level and/or practicing quality improvement in their professional career or entrepreneurial endeavor;

PEO2: Competent in generating new ideas and innovations in Information Technology with more emphasis on Technopreneurship, management, IT solutions and the likes through research collaborations; and

PEO3:
Leading practicing professionals in the field of Information Technology who can contribute significantly to human development, socio-economic transformation, and patriotic initiatives.

Program Outcomes:

a: Apply knowledge of computing, science, and mathematics in solving computing/IT-related problems

II. Course Outcome:

Course Outcomes (CO)	Program Outcome (PO)												
	a	b	c	d	e	f	g	h	i	j	k	l	m
CO1: Students should be able to define and explain the key concepts and techniques related to Decision Support Systems (DSS), including business analytics, business intelligence, data mining, machine learning, predictive analytics, text mining, and deep learning, and demonstrate how these tools are used to support decision-making in organizations.	E	E	E	E	I		I						E
CO2: Students should be able to implement and evaluate various machine learning, text mining, and deep learning models and algorithms to develop and enhance Decision Support System (DSS) technologies, based on the concepts and techniques learned in the course.	D	D	D	E	D		I						D
CO3: Students should be able to create and present a prototype Decision Support System (DSS) solution that integrates appropriate analytics or machine learning techniques	D				D								D

III. Course Outline:

through critical and creative thinking.

b: Use current best practices and standards in solving complex computing/IT-related problems and requirements;

c: Analyze complex computing/IT-related problems by applying analytical and quantitative reasoning; and define the computing requirements appropriate to its solution;

d: Identify and analyze user needs and take them into account in the selection, creation, evaluation and administration of computer based-systems;

e: Design creatively, implement and evaluate different computer-based systems, processes, components, or programs to meet desired needs and requirements under various constraints;

f: Integrate effectively the IT-based solutions into the user environment with appropriate consideration for public health and safety, cultural, societal, and environmental concerns;

g: Select, adapt and apply appropriate techniques, resources, skills, and modern computing tools to complex

Allotted Time	Course Outcomes (CO)	Intended Learning Outcomes (ILO)	Topic/s	Suggested Readings	Teaching-Learning Activities	Assessment Tasks/Tools	Grading Criteria	Remarks
Week 1 5 hours Aug 11-15, 2025			• Class rules/regulations • Attendance Policy • Grading System • Removal/ Special Examinations • Cheating Policy • Submission of Requirements =========== • Leveling of Students Skills and background					
Week 2 & 3 **(10 hours)** Aug 18-29, 2025	**CO1**	1. Understand the need for computerized support of managerial decision making 2. Understand the development of systems for providing decision-making support 3. Understand the different types of analytics 4. Understand the concepts of AI and see selected applications	**I.** Overview of Business Intelligence, analytics, data Science and AI a) The Business Environment and Evolving Needs for Decision Support and Analytics b) The Decision-Making Processes and Computerized Decision Support Framework c) Analytics Overview d) Artificial Intelligence: Concepts, drivers, major Technologies and business applications	-Reference Textbook -Web references	Lecture discussions	-Quiz	Quiz Score	
Week 4 **(5 hours)** Sep 1 - 5, 2025	**CO1**	1. Define data mining as an enabling technology for business analytics 2. Learn the standardized data mining process 3. Learn different methods and algorithms of data mining	II. The Data Mining Process, Methods and Algorithms	-Reference Textbook -Web references	Lecture discussions Readings Assignments	-Quiz -Assignments	Quiz Score	

computing activities, with an understanding of the limitations;

h: Function effectively as individual, or work collaboratively and respectfully as a member or leader in diverse development teams and in multidisciplinary and/or multicultural settings;.

i: Assist in the creation of an effective IT project plan;

j: Communicate effectively in both oral and in written form by being able to deliver and comprehend instructions clearly; and present persuasively to diverse audience the complex computing / IT-related ideas and perspectives;

k: Assess local and global impact of computing information technology on individuals, organizations, and society;

l: Act in recognition of professional, ethical, legal, security and social responsibilities in the utilization of information technology;

m: Recognize the need to engage in independent learning and be at pace with the latest developments in a specialized field in IT, with emphasis on Database Management and

			PRELIM EXAM						
Week 5 - 7 **(15 hours)** Sep 8 - 26, 2025	**CO2**	1. Learn the basics and different types of Machine learning techniques for predictive analytics 2. Apply the different Machine learning techniques to a select datasets	III. Introduction to Machine Learning Techniques for Predictive Analytics a. Neural Networks b. Support Vector Machines c. Nearest Neighbor Method for Prediction d. Naïve Bayes Method for Classification e. Bayesian Network f. Ensemble Modeling	-Reference Textbook -Web references	Lecture discussions and Laboratory Exercises Output Presentation	-Quiz -Laboratory Exercises	Quiz Score Rubric		
Week 8 **(5 hours)** Sep 29 – Oct 1, 2025	**CO2**	1. Learn the basic concepts of deep learning 2. Apply and use the different deep learning techniques into a select dataset (Shallow Network and Deep Neural Net)	IV. Deep Learning and Cognitive Computing a. Introduction b. Basics of Shallow Network c. Process of Developing a Neural Network d. Deep Neural Network	-Reference Textbook -Web references	Lecture discussions and Laboratory Exercises Output Presentation	-Quiz -Laboratory Exercises	Quiz Score Rubric		
Week 9 Oct 2 - 8, 2025			**MIDTERM EXAM**						

Information System; Network Design and Administration; and Computer Vision and Image processing for continual development as a computing professional; n: Participate in generation of new knowledge; or in research and development projects aligned to local and national development agenda or goals with the end view of contributing to the local and national economy; and o: Preserve and Promote "Filipino historical and cultural heritage".	**Week 10 & 11** **(10hours)** Oct 9 - 23, 2025	**CO2**	1. Learn the basic concepts of Convolutional Neural Network (CNN) and Recurrent Neural Network(RNN) 2. Apply and use CNN and RNN to a select dataset	Deep Learning(Cont) e. Convolutional Neural Network f. Recurrent Networks and Long-Short Term Memory	-Reference Textbook -Web references	Lecture discussions and Laboratory Exercises Output Presentation	-Quiz -Laboratory Exercises	Quiz Score Rubric			
	Week 12 & 13 **(10hours)** Oct 24 – Nov 2, 2025	**CO2**	1. describe text analytics, text mining and the different method for text mining 2. Apply and use the different techniques for text analytics to a select dataset	V.Text Mining and Sentiment Analysis a) Text Analytics b) Text Mining c) Overview of Natural Language Processing d) Text Mining Process	-Reference Textbook -Web references	Lecture discussions and Laboratory Exercises Output Presentation	-Quiz -Laboratory Exercises	Quiz Score Rubric			
	Nov 3 - 7, 2025			**Prefinal Exam**							
	Week 15 & 16 **(10hours)** Nov 10 - 21, 2025	**CO2**	1. describe sentiment Analysis, 2. Apply and use the different techniques for sentiment Analysis	e) Sentiment Analysis	-Reference Textbook -Web references	Lecture discussions and Laboratory Exercises Output Presentation	-Quiz -Laboratory Exercises	Quiz Score Rubric			

Week 17 **(5 hours)** Nov 24 – Dec 5, 2025	**CO1, CO2, CO3**		**Final Project Presentation**		Project Presentation and discusison		Rubric	
Dec 8-12, 2025			**FINAL Examinations**					

IV. Course Requirements:

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1. Class attendance and participation policy:
- Attendance refer to Student Handbook
- Expected online classroom behavior
  a. Students are expected to address everyone with respect in every online media class.
  b. Students should also address the Instructor properly. Student may send a message a public or private message for any further clarification in the lessons/topic and in a polite manner.
- Ground Rules for participation in discussions or activities.
  - Only one student may talk at a time. As much as possible, student should turn on their camera during online class.

2. Course Readings/Materials:
(a)Titles, authors, and editions of textbooks and other materials, required and recommended

[1] R. Sharda et. al, Analytics, Data Science, & Artificial Intelligence: Systems for Decision Support, 11th Edition, Pearson Education Limited, 2021.
[2] V. Sauter, Decision Support Systems for Business Intelligence , 2nd Edition, John Wiley & Sons, 2010
[3] DJ Power et. Al, Decision Support Analytics and Business Intelligence, 3rd Edition, Business Expert Press, LLC, 2017
[4]A. Kumar, Learning Predictive Analytics with Python, Packt Publishing, 2016
[5] A. Fuentes, Hands-on Predictive analytics with Python, Packt Publishing

(b) Supplies needed (calculators and programming software)
Python IDE/Anaconda/Jupyter Notebook/ Orange or any integrated development software for programming

(c)URLs for online resources

[6] https://www.datacamp.com/
[7] https://www.kaggle.com/
[8] http://machinelearningmastery.com/
[9] http://towardsdatascience.com/

3. Assignments, Assessment, and Evaluation

(a) Policy concerning assignments

- Programming assignments are given for all lessons. Late submission will result to deduction of points.

(b) Policy concerning make-up exams

- Refer to student handbook. Unless the reason is valid like illness or any emergencies, no make-up or special exam shall be given.

(c) Policy concerning late submission of final projects

- Depending on the reasons, deduct some points or not accepted.

(d) List of requirements that will impact the final grade

Periodic Exams (Prelim, Midterm, Prefinals and Final Exams), Phased Presentation of Project (Prelim, Midterm, Pre-finals and Finals)

(e)Description in detail of grading processes and criteria (how many quizzes, tests, papers; weighting of each; amount of homework, etc.) or the GRADING POLICY

Grading System: (Passing Percentage is 70%)

Midterm Grade

Lecture (50%)		*Laboratory (50%)*	
Midterm Exam	-40%	Lab Assignments	- 70%
Quiz/Prelim (20%/20%)	-40%		
Class Performance	-20%	Midterm Project	- 30%
	100%		100%

Final Term Grade

Lecture (50%)		*Laboratory (50%)*	
Final Exam	-35%	Lab Assignments	- 70%
Quiz/Pre-Final (15%/20%)	-35%		
Performance Innovative Task	-20%	Final Project	- 30%
Class Performance	-10%		100%
	100%		

Final Grade Computation

Final Grade = ½ Midterm Grade + ½ Final Term Grade

(f) Rubrics

a) Laboratory Exercises (Individual Grading)

1) Approved Dataset loaded and split (Train/Test) (10 pts)
2) Model trained properly (10 pts)
3) Model makes predictions (10 pts)
4) Model evaluated (Accuracy or Confusion Matrix shown) (10 pts)
5) New input tested (5 pts)
6) New input tested and model predicts correctly (5 pts)

b) Final Project

The final project will be assessed based on the **technical implementation, presentation clarity, system functionality, and documentation quality**. The following rubric provides a detailed breakdown of the evaluation criteria:

Criteria	Excellent (10 pts)	Good (7-9 pts)	Satisfactory (4-6 pts)	Needs Improvement (1-3 pts)	Not Submitted (0 pts)
Project Implementation & Functionality *(Does the system work as expected? Are ML models properly implemented?)*	The system is fully functional, accurately processes input, and produces correct predictions. ML models are implemented efficiently.	The system works well with only minor bugs. ML models are implemented correctly but may require minor improvements.	The system runs but has some issues affecting functionality. ML models are used but may lack optimization.	The system has major functionality issues, incorrect implementation of ML models, or incomplete features.	No working system submitted.
Machine Learning Model Selection & Performance *(Are appropriate models used? Is performance well-evaluated?)*	The best ML models were selected, well-justified, and optimized for accuracy. Performance metrics (accuracy, precision, recall, F1-score) are clearly presented and interpreted.	Appropriate ML models were used with good performance but minor tuning needed. Metrics are presented but not deeply analyzed.	ML models were selected but not well-optimized. Performance evaluation is basic or lacks interpretation.	Incorrect or inappropriate ML models were used, or performance analysis is missing.	No ML models were used or evaluated.
Data Preprocessing & Feature Engineering *(Was the dataset properly cleaned and prepared?)*	Data preprocessing steps (cleaning, encoding, normalization, feature selection) were well-applied, leading to improved model performance.	Most data preprocessing steps were applied correctly, but minor improvements needed.	Some preprocessing steps were done but may have errors or omissions affecting model performance.	Poor or minimal preprocessing, leading to low-quality input for ML models.	No data preprocessing applied.
User Interface & System Design *(Is the system easy to use? Does the UI allow input and display predictions?)*	The UI is well-designed, user-friendly, and provides clear input/output interaction with the model.	The UI is functional and mostly user-friendly, but could be improved in design and usability.	The UI is basic but works. Some interaction issues exist.	The UI is incomplete, difficult to use, or does not properly interact with the model.	No user interface implemented.
Presentation Clarity & Demonstration *(Was the project clearly explained and demonstrated?)*	Presentation was well-structured, engaging, and covered all key points. System demo was smooth and effective.	Presentation was clear, but some minor points were missing or not well-explained. Demo was mostly successful.	Presentation was somewhat unclear or lacked organization. Demo had issues but was partially functional.	Presentation was difficult to follow, unstructured, or the demo did not work.	No presentation or demo provided.
Documentation Quality *(Was the final report well-written, detailed, and properly formatted?)*	Documentation is complete, well-structured, with clear explanations, diagrams, and references.	Documentation is mostly complete but could use minor improvements in clarity or depth.	Documentation is present but lacks detail, has formatting issues, or is missing key sections.	Documentation is incomplete, poorly written, or lacks clarity.	No documentation submitted.
Problem-Solving & Justification *(Did the student justify choices made in ML techniques and system design?)*	All decisions (model selection, preprocessing, UI design) are well-justified with strong reasoning.	Most decisions are explained, but some justifications lack depth.	Some decisions are justified, but explanations are weak or unclear.	Few to no justifications provided for decisions made in the project.	No reasoning or justification provided.

(g) Date and time of Midterm and Final Exam – refer to the Academic Calendar for SY 2025-2026

4. Use of USTEP, PRISMS and FB group messenger in class to distribute course materials, to communicate and collaborate online, to post grades, to submit assignments, and to give you online quizzes and surveys.

Disclaimer:
Every attempt is made to provide a complete syllabus that provides an accurate overview of the subject. However, circumstances and events make it necessary for the instructor to modify the syllabus during the semester. This may depend, in part, on the progress, needs, and experiences of the students.

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